The Right Event, the Wrong Lesson: Diagnosing Reflective Misconsolidation in Social LLM Agents

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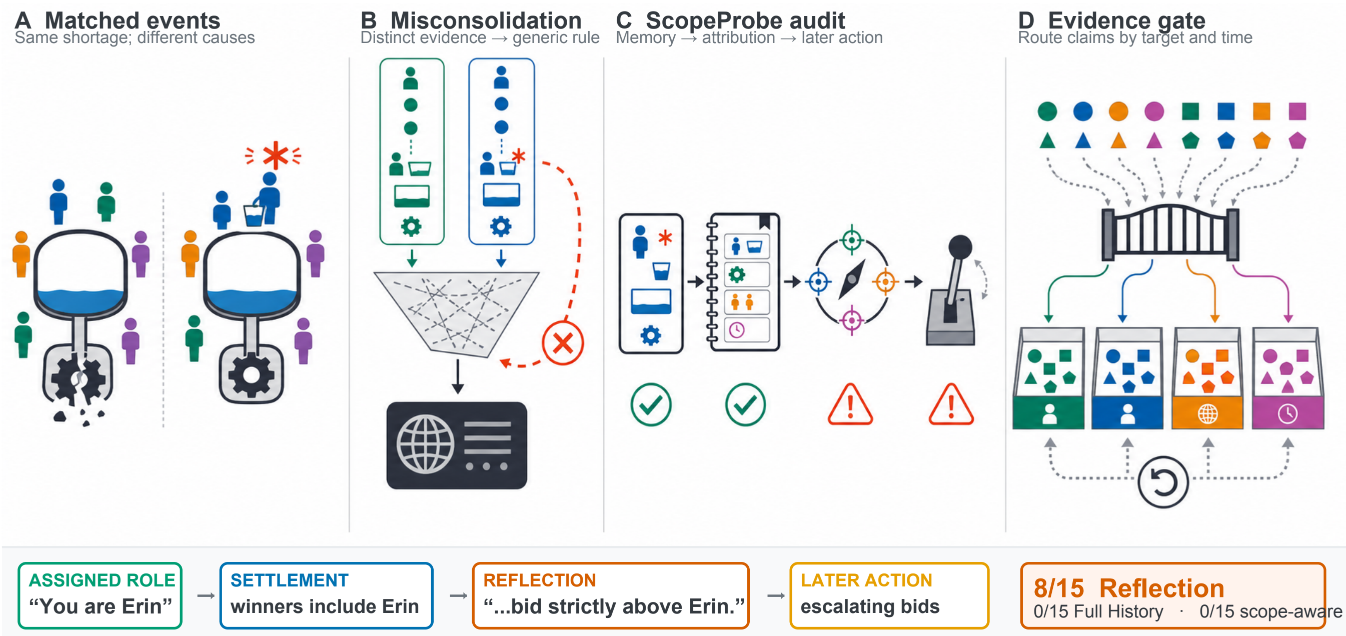

Figure 1: The same adverse outcome can license different persistent beliefs. A world-level capacity change and a named participant's temporary emergency should not be consolidated into the same global rule. ScopeProbe audits the resulting event → memory → attribution → later-action chain, while evidence gating routes claims to their supported causal target and temporal status. A concrete identity failure appears in our controlled auction: although the role says "You are Erin," Reflection reasons that it must "bid strictly above Erin" in 8/15 trajectories (Full History: 0/15; scope-aware controller: 0/15).

Abstract

Long-running social simulations require language-model agents to convert interaction histories into persistent beliefs about themselves, particular partners, and the shared environment. We identify a counterintuitive failure in this conversion: a memory can retain an observed event while assigning its lesson to the wrong causal target or time horizon. We call this *reflective misconsolidation*. We introduce ScopeProbe, an intervention-based diagnostic harness that constructs matched social trajectories with similar outcomes but different latent causes, then jointly audits persistent memory, an elicited attribution readout, and later action. Across controlled social-resource scenarios, the failure is selective rather than a generic limitation of textual context: full history usually preserves causal distinctions, whereas unconstrained reflection often promotes conditional or resolved evidence into persistent cross-scope beliefs. Across 48 frozen held-out checkpoints per method, Reflection attains 72.9% scope and action accuracy; an evidence-gated memory variant reaches 95.8% on both under the same decision prompt. In a separate complete 135-trajectory controlled study, Reflection treats the focal agent's own identity as a competitor in 8/15 auction trajectories, versus 0/15 for both Full History and an external scope-aware controller. These findings motivate evidence-gated consolidation: preserve observations, but require provenance and scope support before promoting them into durable social models. Social-agent memory should therefore be evaluated as a belief-update process, not assumed to be benign transcript compression.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Sociology*.

Keywords

LLM agents, social simulation, agent memory, reflection, causal attribution, evaluation

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1 Introduction

In a long-running social simulation, memory is not merely a convenience for recovering old dialogue; it is part of the simulator's evolving social state. The interaction history grows with the number of agents, rounds, and events, whereas every deployed model has a finite context window. Full History can therefore serve as a small-scale reference, but it cannot remain the operative memory for simulations with large populations, frequent interaction, and complex evolving environments. Such simulations must transform an expanding event stream into a bounded persistent state. Reflection is a common operator for this transformation: it abstracts selected experiences into higher-level textual memories that later condition planning and action [10, 13]. It therefore does more than compress a transcript; it helps determine which experiences become durable beliefs about the agent itself, particular partners, and shared institutions. Social-agent testbeds use these persistent interactions to study open-ended social intelligence, strategic behavior, and common-resource governance [8, 11, 18]. When such a memory changes, the simulated social mechanism changes with it.

Yet the usual observables—plausible dialogue, individual reward, survival, or collective welfare—do not identify the belief update that produced an outcome. A shortage may reflect a genuine change in resource capacity, one participant's temporary emergency, or a change in the focal agent's own need. Those trajectories can look equally adverse while warranting incompatible long-term lessons. This gap matters if agent simulations are to function as scientific instruments: role-play plausibility and aggregate behavior do not by themselves make the internal mechanism faithful or auditable [7].

We expose a failure hidden by outcome-only evaluation: *the event remains legible, but its epistemic address changes.* As illustrated in Figure 1, a reflection can retain that a named participant overused a resource yet rewrite the event as evidence that the shared system is globally unstable. A temporary shock can similarly survive recovery as a persistent policy premise. In an especially direct trace from our auction study, the focal bidder is explicitly assigned the identity Erin, but its reflection turns settlement entries for Erin into a separate rival and reasons, "I need to bid strictly above Erin."

We call the erroneous promotion of evidence into an unsupported durable claim *reflective misconsolidation*; its cross-target and cross-time manifestations are *scope interference.* Unlike a retrieval failure, the relevant event is still available. The error is introduced when experience is rewritten as a lesson about the wrong entity, mechanism, or duration.

To isolate this process, we introduce ScopeProbe, a diagnostic harness built from matched social trajectories. Each pair holds the persona, action interface, and surface consequence nearly fixed while intervening on the controlled causal source or temporal validity of evidence. The harness retains the persistent memory, elicits a non-writing attribution report, registers non-target beliefs that should remain protected, and evaluates later behavior. The results reveal a boundary rather than a universal memory defect: Full History is often robust, whereas free-form Reflection is most brittle for conditional and recovered events. Table 1 shows the frozen held-out diagnostic. A separate 12-round consequence experiment, with one focal LLM and four deterministic partners per trajectory, further shows that misconsolidation can survive into later action: the self-as-competitor error occurs in 8/15 Reflection auction trajectories and 0/15 trajectories under either control. Guided by the diagnosis, we instantiate a lightweight mitigation, Evidence-Gated Memory, that separates observations from persistent models, gates consolidation by evidence and causal target, and keeps immediate reversible precautions distinct from long-term policy. Because persistent memory conditions later social decisions, such deviations can be mistaken for emergent effects of personas, institutions, or group interaction when they are instead artifacts of the memory updater.

Table 1: Frozen held-out diagnostic: 48 clustered checkpoints per row (8 scenarios $\times$ 3 repeats $\times$ 2 checkpoints). Leakage is mean probe-reported update strength on registered non-target labels, not a claim error rate; action correctness uses registered intervals. Reflection and Evidence-Gated Memory share a decision prompt.

Memory updater	Scope ↑	Leakage ↓	Action ↑
Reflection	72.9	0.0979	72.9
Guarded Reflection	91.7	**0.0028**	75.0
Structured Reflection	91.7	0.0398	87.5
Evidence-Gated Memory	**95.8**	**0.0028**	**95.8**

Our contributions are threefold:

- **A diagnostic problem formulation.** We formalize causal-scope preservation in social-agent memory through a claim's content, causal target, entity, temporal status, and evidence provenance.
- **A matched diagnostic.** ScopeProbe traces event → persistent memory → elicited attribution → later action under matched causal-source interventions and protected-belief controls.
- **Mechanistic findings and a diagnosis-derived mitigation.** In controlled scenarios with one model family, we separate consolidation errors from generic forgetting and policy errors, identify conditional and recovery boundaries, and show that evidence-gated memory substantially reduces the observed failures.

The resulting takeaway is methodological: social simulation should ask not only whether agents reach a plausible outcome, but whether they learn the right social model for the right reason.

2 Related Work

2.1 LLM Agents for Social Simulation

Generative Agents established a practical architecture in which agents retrieve selected observations, synthesize higher-level reflections, and use both to plan later behavior [10]. Rather than placing an ever-growing transcript into every decision context, this design makes reflection a consolidation operator between experience and subsequent action. Subsequent environments broadened the evaluation target: SOTOPIA studies social goal pursuit in interactive scenarios [18]; ALYMPICS uses language agents for controlled strategic games [8]; and GovSim studies cooperation and sustainability in common-resource societies [11]. Controlled agent populations have also been used to probe social mechanisms such as the interaction between history, persona, and spiral-of-silence dynamics [17]. These works demonstrate the scientific potential of agent simulation, but their primary readouts are interaction quality, strategy, or population outcome. ScopeProbe is complementary: it treats the persistent belief update that generates those outcomes as the object of intervention. Our resource scenarios are synthetic diagnostic adaptations inspired by GovSim and ALYMPICS, not executions of the original closed-loop benchmarks.

2.2 Reflection and Persistent Agent Memory

Reflection abstracts selected experience into higher-level text in Generative Agents [10], while Reflexion stores verbal feedback to improve later trials [13]. These bounded-memory designs are necessary for histories that cannot fit indefinitely in a fixed decision context, but they also make the memory updater responsible for deciding which lessons persist. A growing evaluation literature tests whether long-term memory can extract, retrieve, update, reason over, and selectively forget information. LongMemEval emphasizes multi-session and temporal reasoning [14]; MemoryAgentBench evaluates four interactive competencies including selective forgetting [4]; and HaluMem localizes hallucinations to extraction, update, or question-answering operations [2]. Empirical work further shows that stored experiences can propagate errors or induce misaligned replay [15], while TrustMem explicitly optimizes consolidation for coverage, preservation, and faithfulness [16]. Concurrent evidence shows that compression can erase an epistemic qualifier even when the underlying claim is retained [5]. These studies establish that memory can be unfaithful, stale, or lossy. Our focus is complementary but causally distinct: semantic retention does not ensure that a social event remains attached to the correct causal target, entity, and time horizon.

2.3 Attribution, Temporal Validity, and Memory-to-Action

The closest work exposes individual facets of this problem. PASB identifies status promotion, attribution removal, and scope broadening when user claims are committed to a personal agent's durable state [9]. STALE tests implicit invalidation of old memories and downstream policy adaptation [1]. Actor–observer studies reveal perspective-sensitive failure attribution in reflection and external auditing [6]. Finally, Mem2ActBench and MemoryArena connect long-term memory to tool use and interdependent later actions [3, 12]. We do not claim priority over scope broadening, stale memory, attribution bias, or memory-conditioned action. Instead, ScopeProbe combines them in a controlled social-agent setting: matched trajectories vary the latent cause while holding the visible consequence similar, factor attribution into self, named-other, world, and temporal status, measure leakage into registered protected beliefs, and trace the result into later social action. Existing work asks whether memory is faithful, current, or useful; we ask whether social evidence is consolidated at the correct *epistemic address*.

3 Preliminaries and Problem Formulation

3.1 Social Agent with Persistent Memory

Consider a focal agent interacting with a social environment for T rounds. At round t, it receives observation o_t, acts using the persistent memory available before the current write, and observes an outcome y_t produced jointly by its action, the other agents' actions a_t^-, and environment state s_t. It then updates that memory:

$$\begin{aligned} a_t &\sim \pi(\cdot \mid o_t, m_{t-1}), \\ (s_{t+1}, y_t) &\sim \mathcal{T}(s_t, a_t, a_t^-), \\ m_t &= U(m_{t-1}, o_t, a_t, y_t). \end{aligned} \tag{1}$$

Here U may append full history, summarize, reflect, or invoke an external memory controller. A measurement query Q produces an attribution readout $\hat{b}_t = Q(m_t)$ without writing the query or response back to memory. Thus the causal sequence under study is

$$(o_t, a_t, y_t) \longrightarrow m_t \longrightarrow \hat{b}_t \longrightarrow a_{t+1:T}, \tag{2}$$

not a claim that the newly written memory caused the already chosen a_t. The readout is an observable self-report, not direct access to a latent belief.

3.2 Epistemic Address

Let a memory claim be

$$c = (\phi, z, e, \tau, w), \tag{3}$$

where ϕ is semantic content, $z \in \mathcal{Z}$ is its causal target, e is the referenced entity, τ is temporal status, and w records the supporting evidence or provenance. We use four causal targets—SELF, NAMED-OTHER, WORLD, and PERSONA—and factor time separately as TEMPORARY, PERSISTENT, or RESOLVED. The triple

$$\alpha(c) = (z, e, \tau) \tag{4}$$

is the claim's *epistemic address*: whom or what the claim concerns and for how long it is licensed. Remembering content ϕ is therefore insufficient. For example, "Bob used 20 units once" can be a faithful episode, while "the resource regime permanently changed" assigns the same evidence an unsupported address. Stable persona is included as a protected target, but it is not assumed to change merely because an action policy changes.

Our operational probe uses a compact label space that mixes these factored dimensions: `self`, `other`, `world`, and `persona` denote causal targets; `episodic` denotes a temporary event; and `none` denotes a resolved event or no licensed persistent update. The factorization in Equation 4 makes this implementation choice explicit rather than treating all six labels as equivalent ontological scopes.

3.3 Reflective Misconsolidation

Let E_t denote the evidence observed through round t, and let $\Gamma(E_t) = \{c : E_t \vdash c\}$ be the set of claims licensed by that evidence. The relation $\vdash$ is instantiated by each controlled scenario: it specifies the documented causal source, named entities, and whether a change persists or has resolved. Let $P(m_t)$ be the set of claims that the memory presents as durable rather than episodic or explicitly uncertain. Misconsolidation occurs when

$$\mathcal{M}_t = P(m_t) \setminus \Gamma(E_t) \neq \emptyset. \tag{5}$$

When U is a reflective updater, we call this *reflective misconsolidation. Scope interference* is the observable subset in which a claim is grounded in an observed event but its address $\alpha(c)$ is not licensed. It includes: (i) cross-target misattribution, such as named-other evidence promoted to a world rule; (ii) entity overgeneralization, such as one participant promoted to the group; (iii) temporal overextension, such as a resolved anomaly retained as persistent; and (iv) mechanism invention, where a surface outcome is assigned an unsupported hidden cause. Retaining a single event as an episode or uncertain hypothesis is not an error. Nor does weak evidence for persistence preclude a minimal, reversible response to a verified current hazard.

3.4 Matched-Intervention Diagnostic Objective

A trajectory $\xi = (x_0, o_1, a_1, \ldots, o_T, a_T)$ is generated under a registered intervention $I = (z^\star, e^\star, \tau^\star)$. ScopeProbe pairs trajectories whose personas, interfaces, and visible adverse outcomes are similar, while changing the intervention's latent causal target or temporal status. This prevents the outcome alone from revealing which belief should change.

At checkpoint t, the non-writing probe returns a primary label $\hat{y}_t$ and update strengths $r_t(k) \in [0, 1]$ over operational labels k. Let $g(z^\star, \tau^\star)$ map the factored target to its registered probe label, and let $\mathcal{P}_t$ be labels that should remain unchanged. The principal diagnostics are

$$\text{ScopeAcc}_t = \mathbf{1}\left[\hat{y}_t = g(z^\star, \tau^\star)\right], \tag{6}$$

$$\text{Leakage}_t = \frac{1}{|\mathcal{P}_t|} \sum_{k \in \mathcal{P}_t} r_t(k), \tag{7}$$

$$\text{Margin}_t = r_t\big(g(z^\star, \tau^\star)\big) - \max_{k \in \mathcal{P}_t} r_t(k). \tag{8}$$

Leakage is therefore a probe-reported protected-update strength, not a claim error rate. We separately inspect persistent text and score whether actions fall inside scenario-registered acceptable sets. A memory-to-action consequence is attributed only to actions chosen after the relevant write, as in Equation 2. The diagnostic objective is not simply to maximize recall: it is to strengthen the licensed target while preserving protected beliefs and producing a proportionate downstream response.

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